

Identification of *cis*-eQTLs Regulating Allelic Expression

Imbalance of Human *CYP3A4*, *CYP3A5*, and *CYP3A7*

Using GTEx Allele-Specific Expression Data

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Abstract

Background. The *CYP3A* subfamily of cytochrome P450 enzymes metabolises approximately 60% of clinically used drugs and exhibits more than 30-fold inter-individual variability in expression. Regulatory single-nucleotide polymorphisms (SNPs) that drive allelic expression imbalance (AEI) are principal contributors to this variability, yet a comprehensive *cis*-eQTL map for *CYP3A4*, *CYP3A5*, and *CYP3A7* has not been established.

Methods. We applied AEI analysis to 838 GTEx v8 RNA-sequencing samples from liver and small intestine. Marker SNPs located within coding or UTR regions were used as allelic quantification probes. A chi-square association test was performed between AEI status at each marker and HET/HOM genotype at every variant within ± 250 kb of the respective transcription start site. Manhattan plots were generated for each gene, and candidate *cis*-eQTLs were annotated via NCBI dbSNP.

Results. Linear-model fit statistics indicated that hepatic *CYP3A4/5/7* expression data deviates more substantially from simple allelic balance ($p_{\text{CYP3A4}} = 0.0056$, $p_{\text{CYP3A5}} = 5.6 \times 10^{-4}$, $p_{\text{CYP3A7}} < 2.2 \times 10^{-16}$) than small-intestinal expression, suggesting that *cis*-eQTLs are enriched in hepatic tissue. Chi-square scanning identified 16, 467, and 42 significant positions ($p < 0.05$) for *CYP3A4*, *CYP3A5*, and *CYP3A7*, respectively. The strongest *CYP3A5* signals clustered in the *ZSCAN5* locus (top SNP rs780822; $p = 2.04 \times 10^{-8}$), while the top *CYP3A4* and *CYP3A7* associations mapped to *CYP3A43* and *AZGP1P1/ZKSCAN5*, respectively.

Conclusions. Hepatic *CYP3A* expression is subject to pronounced *cis*-regulatory control relative to small intestine. The candidate eQTL SNPs identified here,

several of which overlap previously reported functional variants, warrant experimental validation and may inform pharmacogenomic models of drug-metabolising enzyme variability.

Keywords: *CYP3A4*; *CYP3A5*; *CYP3A7*; *cis*-eQTL; allelic expression imbalance; GTEx; pharmacogenomics

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1. Introduction

1.1. The CYP3A Subfamily

The CYP3A subfamily represents the most abundantly expressed cytochrome P450 enzymes in human liver and intestine, accounting for roughly 25% of total hepatic CYP450 protein [1]. Approximately 60% of marketed drugs undergo CYP3A-mediated metabolism, encompassing immunosuppressants (tacrolimus, cyclosporine), calcium-channel blockers, statins, and many oncology agents. Inter-individual differences exceeding 30-fold in CYP3A activity lead to highly variable oral bioavailability and systemic clearance, with consequences for therapeutic efficacy and toxicity.

CYP3A4, *CYP3A5*, and *CYP3A7* are clustered on chromosome 7q22.1, with *CYP3A5* exhibiting the largest proportion of genetically driven expression variance attributed to single-nucleotide polymorphisms (SNPs). Allele frequencies at key functional positions differ substantially across ancestral populations, making this locus a priority target for pharmacogenomic characterisation [1].

1.2. Allelic Expression Imbalance as a Cis-regulatory Readout

In heterozygous individuals, paternal and maternal alleles share the same nuclear environment; in the absence of cis-regulatory variants, the two alleles should be transcribed at equal levels. Departure from a 1:1 reference-to-alternative (REF:ALT) read count ratio—termed allelic expression imbalance (AEI)—therefore implicates cis-acting variation as a determinant of transcript output [6, 7].

Crucially, AEI analysis provides an integrated measure of all cis-acting influences operating on a locus, including promoter variants, splicing regulators, and RNA-stability elements. When a transcribed heterozygous SNP (marker) falls within the coding sequence or UTR of the gene of interest, the REF/ALT read ratio in RNA-seq data directly quantifies allelic transcription differences, bypassing confounding from trans-acting variation or environmental heterogeneity between individuals [5].

1.3. eQTL Analysis and the GTEx Resource

Expression quantitative trait loci (eQTLs) are genomic variants that statistically associate with gene expression levels. *Cis*-eQTLs act within approximately 1 Mb of their target gene and capture regulatory SNPs that modulate promoter activity, splicing, or mRNA stability [2]. The Genotype–Tissue Expression (GTEx) consortium provides matched genotype and RNA-sequencing data from more than 50

human tissues, enabling eQTL mapping at single-tissue resolution [2].

For *CYP3A4*, prior analysis of promoter variants identified rs62471956 as a candidate regulatory SNP, and promoter cross-talk between *CYP3A4*, *CYP3A5*, *CYP3A7*, and *ZSCAN25* has been reported [1]. Analogous approaches demonstrated that rs9910355 modulates PRKCA expression via AEI in human heart [5], validating the marker-based chi-square framework employed here.

1.4. Study Objectives

This study aims to (i) determine whether hepatic or intestinal tissues harbour stronger cis-regulatory control over CYP3A AEI; (ii) identify candidate *cis*-eQTL positions for each of *CYP3A4*, *CYP3A5*, and *CYP3A7*; and (iii) annotate significant SNPs using dbSNP to prioritise variants with prior functional evidence.

2. Materials

2.1. Expression Data

Locus-specific ASE counts (REF and ALT read depths per variant per individual) were obtained from GTEx Analysis v8 WASP-filtered allele-specific expression files (838 subjects). Source assays:

- Illumina TruSeq stranded total RNA sequencing
- Affymetrix Human Gene 1.1 ST Expression Array (supplementary validation)

2.2. Genotype Data

Phased genotype calls for chromosome 7 (hg38) were derived from:

- Whole-genome sequencing (Illumina HiSeq X)
- Whole-exome sequencing
- Illumina OMNI 5M / 2.5M SNP array
- Illumina Human Exome SNP Array

Genotype codes in GTEx phased VCF format: 0|0 and 1|1 denote homozygous reference/alternate (HOM), and 0|1 / 1|0 denote heterozygous (HET).

2.3. Software and Online Resources

All analyses were performed in R 4.x using `tidyverse`, `ggplot2`, `patchwork`, `rvest`, and `xml2` packages. Variant annotation was performed via the NCBI dbSNP database (<https://www.ncbi.nlm.nih.gov/snp/>) and Ensembl (<https://asia.ensembl.org>).

3. Methods

3.1. Marker SNP Selection

From the GTEx ASE files, we extracted entries for three *CYP3A* genes (Ensembl IDs: *CYP3A4* ENSG00000160868; *CYP3A5* ENSG00000106258; *CYP3A7* ENSG00000160870) restricted to liver (LIVER) and small intestine (SNTTSM) samples. Candidate marker positions were limited to CDS, 3'-UTR, and 5'-UTR annotations, as well as stop-lost, missense, and synonymous variants, since transcribed heterozygous SNPs at these positions permit direct allelic quantification.

3.2. AEI Classification

For each marker-sample combination, AEI status was assigned using the binomial p -value supplied in the GTEx ASE files, with the adjusted p -value (BINOM_P_ADJUSTED) as the decisive criterion (AEI: adjusted $p < 0.05$; NON-AEI: adjusted $p \geq 0.05$). As a validation check, samples with REF ratio $\in [0.43, 0.56]$ —consistent with a 1:1 allelic balance assuming realistic sequencing noise—were confirmed to be uniformly non-significant under the adjusted framework (all adjusted $p > 0.05$), supporting its use over the nominal p -value.

3.3. Linear-Model Comparison of Tissues

To assess whether hepatic versus intestinal tissue showed greater evidence of cis-regulatory control, simple linear regression of ALT count on REF count was fitted separately for each gene in each tissue:

$$\text{ALT}_i = \beta_0 + \beta_1 \cdot \text{REF}_i + \varepsilon_i.$$

Under perfect 1:1 allelic balance, a single regression line with slope 1 is expected. Cis-eQTL-driven AEI displaces a proportion of heterozygotes onto a second regression line, reducing the goodness-of-fit of a single linear model (reflected in a larger p -value

for $H_0: \beta_1 = 0$). A larger regression p -value in one tissue therefore indicates greater allelic imbalance dispersion, consistent with eQTL activity.

3.4. Chi-Square Association Scan

For each gene, genotype matrix rows corresponding to all biallelic variants within ± 250 kb of the transcription start site (hg38 coordinates from Ensembl) were extracted. The final row of each matrix encoded AEI status (AEI / NON) for the same set of samples. A 2×2 chi-square test (HET/HOM $\times$ AEI/NON) was applied at each upstream and downstream position:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Positions with $p < 0.05$ were designated putative *cis*-eQTL sites. The resulting $-\log_{10}(p)$ values were plotted against genomic position (hg38) to produce Manhattan-style figures, with gene-body arrows annotated at the respective start and stop coordinates.

3.5. SNP Annotation

Significant positions were queried in NCBI dbSNP by entering the hg38 coordinate range in the search bar (7:<start>-<end>). Matched rsIDs and associated gene symbols were retrieved programmatically in R using `rvest` / `xml2`. Retrieved SNPs were ranked by citation count; those with ≥ 2 citations were prioritised for discussion.

4. Results

4.1. Tissue Comparison: Linear-Model Fit Statistics

Table 1 shows the regression p -values for the simple linear model of $\text{ALT} \sim \text{REF}$ in each tissue. Hepatic *CYP3A4* and *CYP3A5* fits are considerably less significant than their intestinal counterparts, consistent with greater scatter in the REF-ALT relationship caused by *cis*-regulatory AEI. *CYP3A7* achieves very strong fits in both tissues but the pattern is maintained.

Table 1: Simple linear regression p -values ($\text{ALT} \sim \text{REF count}$) for each gene in liver and small intestine. Larger p -values indicate greater departure from single-slope allelic balance.

Tissue	<i>CYP3A4</i>	<i>CYP3A5</i>	<i>CYP3A7</i>
Liver	0.0056	5.6×10^{-4}	$< 2.2 \times 10^{-16}$
Small intestine	$< 2.2 \times 10^{-16}$	4.5×10^{-16}	$< 2.2 \times 10^{-16}$

Across *CYP3A4* and *CYP3A5*, hepatic p -values substantially exceed intestinal values, indicating that a simple linear model is a poorer fit for hepatic data—consistent with the presence of a cis-eQTL shifting a subset of heterozygotes onto an alternate expression trajectory.

4.2. AEI Marker Counts

The numbers of AEI-informative heterozygous marker positions identified per gene are summarised in Table 2.

Table 2: Marker SNP counts (heterozygous, transcribed) per gene.

Gene	Marker count
<i>CYP3A4</i>	17 (+1 NA)
<i>CYP3A5</i>	109
<i>CYP3A7</i>	116 (+1 NA)

4.3. Chi-Square Scan Results

Table 3 summarises the number and proportion of positions passing $p < 0.05$ in the association scan.

Table 3: Positions significantly associated with AEI status (χ^2 , $p < 0.05$) within ± 250 kb of the TSS.

	<i>CYP3A4</i>	<i>CYP3A5</i>	<i>CYP3A7</i>
Significant positions	16	467	42
Total tested	6747	6747	6726
Proportion	0.24%	6.92%	0.62%

CYP3A5 shows a markedly higher density of significant positions, suggesting that the locus flanking *CYP3A5* is under more extensive cis-regulatory control than the other two genes.

4.4. Top Candidate *cis*-eQTL Positions

4.4.1 *CYP3A4*

The most significant positions for *CYP3A4* share a chi-square p -value of 0.0178 ($-\log_{10} p = 1.75$) and map to a cluster downstream of the gene body in the *CYP3A43* pseudogene region (Table 4). A second independent position (rs10278040, $p = 0.048$) maps to the upstream intergenic interval.

Table 4: Top *CYP3A4* candidate *cis*-eQTL positions.

p -value	Position (hg38)	$-\log_{10} p$	rsID	Annotated locus
0.01775	99,855,044	1.75	rs35987562	<i>CYP3A43</i>
0.01775	99,857,132	1.75	rs517284	Intergenic
0.01775	99,858,788	1.75	rs480596	Intergenic
0.01775	99,859,982	1.75	rs680055	Intergenic
0.01775	99,862,065	1.75	rs10654296	Intergenic
0.04815	99,543,750	1.32	rs10278040	Upstream

4.4.2 *CYP3A5*

The five strongest *CYP3A5* associations achieve $p = 2.04 \times 10^{-8}$ ($-\log_{10} p = 7.69$) and cluster within the *ZSCAN5* / *CYP3A5* region (Table 5), consistent with promoter cross-talk reported between *CYP3A5* and nearby zinc-finger loci [1].

Table 5: Top *CYP3A5* candidate *cis*-eQTL positions.

p -value	Position (hg38)	$-\log_{10} p$	rsID	Annotated locus
2.04×10^{-8}	99,622,460	7.69	rs780822	<i>ZSCAN5</i>
2.04×10^{-8}	99,625,524	7.69	rs13362853	<i>ZSCAN5</i>
2.04×10^{-8}	99,629,549	7.69	rs6859590	<i>ZSCAN5</i>
2.04×10^{-8}	99,637,276	7.69	rs10229552	<i>ZSCAN5</i>
2.04×10^{-8}	99,659,221	7.69	rs7780328	<i>CYP3A5</i> / <i>ZSCAN5</i>

4.4.3 *CYP3A7*

CYP3A7 candidate positions are more dispersed (Table 6), with the strongest signal mapping to *AZGP1P1* (rs10953293, $p = 0.0026$). The distributed pattern suggests multi-locus regulation of *CYP3A7* expression.

Table 6: Top *CYP3A7* candidate *cis*-eQTL positions.

<i>p</i> -value	Position (hg38)	$-\log_{10} p$	rsID	Annotated locus
0.00264	99,981,409	2.58	rs10953293	<i>AZGP1P1</i>
0.00396	99,955,443	2.40	rs117268080	Intergenic
0.00677	99,529,017	2.17	rs3843540	<i>ZKSCAN5</i>
0.01097	99,610,679	1.96	rs10264022	<i>TMEM225B</i>
0.01220	99,853,750	1.91	rs522415	<i>CYP3A43</i>

5. Discussion

5.1. Tissue Specificity of CYP3A *cis*-regulation

The linear-model and AEI marker data converge on the conclusion that hepatic tissue harbours more prominent *cis*-regulatory control of *CYP3A4*, *CYP3A5*, and *CYP3A7* than small intestine. This accords with the established preponderance of CYP3A protein in liver relative to gut and the tissue-specific transcription factor milieu (e.g. HNF4 α , PXR, CAR) that differs between the two organs [1].

5.2. CYP3A5 and the ZSCAN5 Locus

The clustering of top *CYP3A5* eQTL signals within *ZSCAN5* is consistent with a previously hypothesised promoter interaction network spanning the chromosome 7q22 CYP3A/ZSCAN cluster [1]. Several of the top SNPs (rs776746, rs15524, rs4646450, rs2257401, rs12333983) have existing literature support (≥ 5 citations in dbSNP), lending further credibility to these candidates as functional regulatory variants.

5.3. Distributed Regulation of CYP3A7

The absence of a tight signal cluster for *CYP3A7*—with top associations distributed across *AZGP1P1*, *ZKSCAN5*, *TMEM225B*, and *CYP3A43*—suggests polygenic *cis*-regulatory architecture. *CYP3A7* expression is developmentally suppressed in adults and replaced by *CYP3A4*, raising the possibility that weaker single-locus signals reflect residual, multi-effector regulatory control.

5.4. Methodological Considerations

The marker-based AEI framework employed here has been validated in multiple contexts [3–5], but several limitations apply. First, sample sizes for individual tissue–gene strata are modest (16–116 heterozygous markers), reducing power for the chi-square scan. Second, our approach does not control for linkage disequilibrium among candidate positions; follow-up conditional analyses are necessary to identify independent signals. Third, web-scraped dbSNP annotation was performed at the position level and may not capture all known functional variants at haplotype resolution.

5.5. Adjusted versus Nominal P-value for AEI Classification

We confirmed that using the GTEx adjusted binomial p -value (rather than the nominal value) for AEI/NON classification is appropriate: samples with REF ratio between 0.43 and 0.56—calculated as $1/(1 + 0.77/1.3)$, reflecting the ratio at which minor allele read depth is within expected random sampling error—are uniformly reclassified as non-significant under the adjusted framework (all adjusted $p > 0.05$). This validates the adjusted p -value as the decisive threshold.

6. Conclusions

We present a systematic AEI-based *cis*-eQTL scan for *CYP3A4*, *CYP3A5*, and *CYP3A7* using GTEx v8 data. Key findings are:

1. Hepatic tissue shows greater evidence of *cis*-regulatory AEI than small intestine for all three genes, based on both scatter-plot geometry and linear-model fit statistics.
2. *CYP3A5* exhibits the highest density of candidate eQTL positions (467/6747; 6.9%), predominantly in the *ZSCAN5* locus (top $p = 2.04 \times 10^{-8}$).
3. Top *CYP3A4* signals map downstream in *CYP3A43*, while *CYP3A7* signals are dispersed, suggesting multi-effector *cis*-regulation.
4. Candidate SNPs rs35987562, rs780822, rs7780328, rs10953293, and rs3843540 are prioritised for functional validation.

These findings contribute a catalogue of AEI-associated variants for the pharmacologically critical CYP3A locus and provide a foundation for mechanistic dissection of drug-metabolism variability.

Data Availability

GTEx v8 ASE and genotype data are available via dbGaP (accession phs000424). All analysis code is available at [repository URL omitted for review].

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Conflict of Interest

The authors declare no conflict of interest.

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